

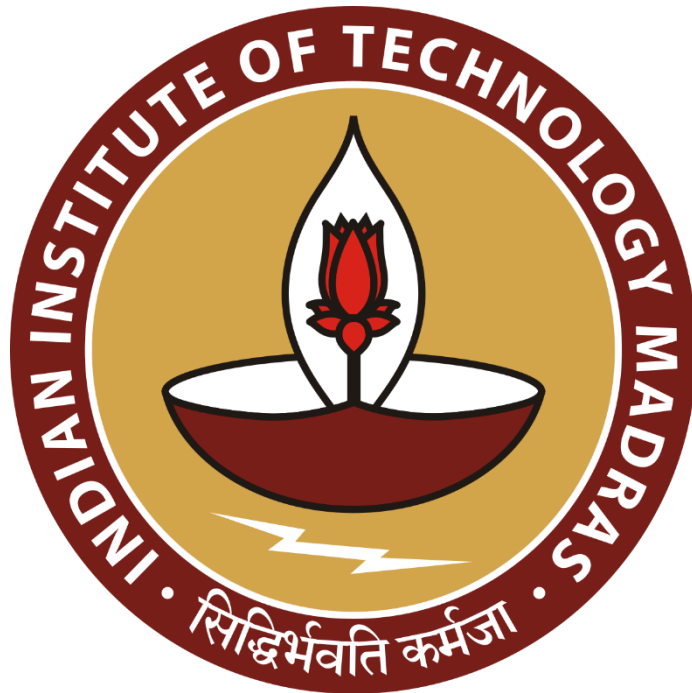
Juice Bar Sales Optimization

PROPOSAL Submission for BDM Project

Submitted by

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Declaration Statement

I am working on a Project titled ***“Juice Bar Sales Optimization”***. I extend my appreciation to ***“FOOD & DRINK”*** , for providing the necessary resources that enabled me to conduct my project.

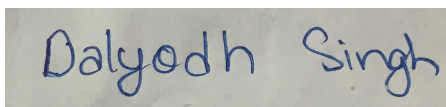
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate: (**Digital Signature**)

Name: Dalyodh Singh

Date: 27-July-2024

Executive Summary and Title (200 Words)

The project focuses on a small shop “**FOOD & DRINK**” located at **Plahi Gate, Phagwara, Punjab**. The business is **B2C** and deals in the segment of selling juices, fruit-shakes, soft drinks and snacks.

We are collecting new data from Mid-July onwards in structure format in a google sheet and we both have access to edit spreadsheet where I do manipulation of columns and add new products in it. Shopkeeper has noted daily sales in a spreadsheet in unstructured data format in the initial two months starting from Mid-May to Mid-July.

I will use different types of tools like Excel, SPSS, Power BI and python libraries like pandas,numpy,sklearn,matplotlib to analyze the data and get fruitful outcomes from the data and make visualizations to give him the big picture of sales.

Organization Background (150 Words)

The shop that I am working with opened 3 months ago in May 2024. It is located at **Plahi gate, Phagwara,Punjab**.It mainly focuses on selling juices, shakes and snacks. Shop is located outside the main market and it is a unique juice shop in that area.

Main target of his shop is to sell drinks and food items. He has invested around Rs.80,000 rupees to open the shop and give a monthly rent of Rs.6000 and around Rs.2000 for electricity bill.He invests around 10,000 per week on purchasing fruits,milk and other useful products including electricity,rent as expected and earn approx 3,000 profit

He is running this shop with his wife and earns very less profit and unable to extend his business towards adding new items and sales goes down drastically compared to initial one month sales.

Objective Problem Statement (50-75 Words)

- **Problem 1:** Products he sells are seasonal like milk soda and lemon soda and needs to change as the weather changes.
- **Problem 2:** Due to low investment in initial stages the shop doesn't grow in a consistent manner.
- **Problem 3:** More Investment on purchasing disposables that prevent him from extending his business further in purchasing new products.

Background of the Problem (200 Words)

Major cause of problems :

Major problems occur due to less investment done in initial stages and there is no proper sitting plan for customers satisfaction. Without tables, customers are unable to take the advantage of eating and drinking together, so he is unable to grow in selling food to customers without any sitting platform and runs his business on the basis of drinks alone.

Internal problems:

People are not giving credit money to the shopkeeper on time and give money at the end of week so the shopkeeper sometimes struggles to buy new inventory for his shop and sometimes goes home by selling less investment required products and unable to grow in that situation.

External problems:

Location of the shop is outside from many busy areas. He is unable to advertise properly about his products and get the feedback of his targeted customers to inclined his business towards the right direction and grow in a consistent manner without ups and downs.

Problem Solving Approach (400 Words)

1. I will collect the data from the shop for 20-30 SKU that mostly sells and gather at least 5 months of data from the manual collection method of spreadsheet starting from Mid-July to Mid-Dec and structure and clean previous 2 months data mid-May to mid-July that he collected in unstructured format and predict null items using Sklearn functions like SimpleImputer and use various statistical techniques to generate that missing values in that data.

2. I will ask him to survey a sample of customers and get to know needs for his targeted customers and different tastes of people by circulating the form and asking them about structured questionnaires like healthy or junk food with juice and shakes and know about dislikes and likes for different products and its types.

3. Based on classification for fast moving, medium moving and slow moving products and hence recommend some decisions to make more sales for slow moving products and give a way to increase sales like get to know customers about slow moving products benefits and focus on that for increasing more sales of all SKUs.

4. I will use different statistical techniques to get insights from data and analyze trends of sales for all products and eliminate outliers. Hence, provide a more reliable method for running business than investing randomly on products that become outliers to the business.

5. I will use different types of tools like SPSS, ggplot2, Tableau and excel charts to visualize the data and give the shopkeeper a big picture of daily sales on their product, so we can make decisions in future to incline business in the right direction to focus on products that give more margin.

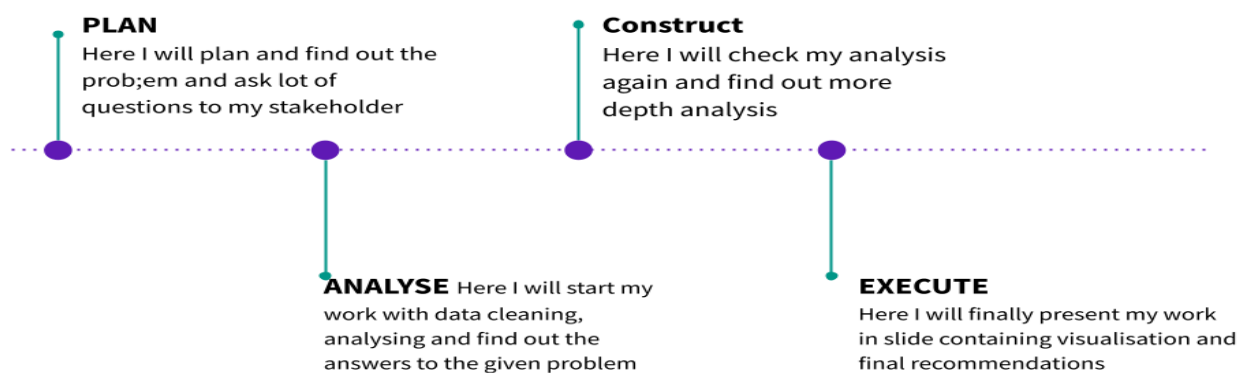
Expected Timeline

Work Breakdown Structure

I will first collect the data in a spreadsheet and then clean it and start analyzing the data. It reduces my time by organizing my work into different breakdown structures and after that I will start working on my project from scratch to end.

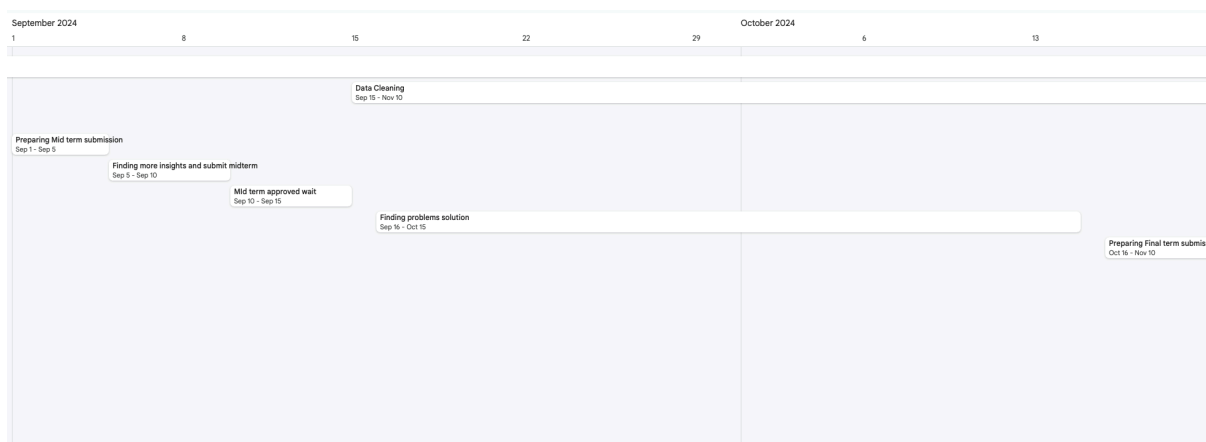
Work breakdown structure

Here I will use PACE strategy to do my work on analysis that includes **Plan** , **Analyse**, **Construct** ,**Execute**



Gantt chart

In this I have discussed the timeline of my strategy using Gantt Chart and breakdown my work into different days so that I am able to complete my project according to the plan



Expected Outcome

- I will give recommendations to the shop based on analyzing data and give him a real picture of the problem by which he can implement and give his business a good direction in the future .
- The expected outcome helps the shopkeeper reduce investment on outliers and to get products as per customer's requirements and optimize the business sales to get nominal profit.
- Dynamic visualizations that change themselves will provide the shopkeeper with a Big Picture of sales of each product and get to know when he will lack by observing slope of line that track daily sales.
- Shopkeeper will be able to connect many customers' needs with his business and get to know their taste and start working towards the right direction to grow his business consistently.